

Reinforcement Learning using EEG signals for Therapeutic Use of Music in Emotion Management on DEAP and DENSE Datasets

Sahil Nagwani Vishal Gautam Prakriti Manhar
Department of Artificial Intelligence
Guided by: Sudhakar Mishra

Abstract—Using music to regulate emotions presents an interesting alternative to medication for managing negative moods, but the problem is that picking the right music manually doesn’t work very well—what helps one person might make another feel worse. In this paper, we develop a Reinforcement Learning system that uses real-time EEG signals to detect someone’s emotional state and then selects music to guide them toward feeling better. We tested three different RL approaches—Q-Learning, SARSA, and Double Q-Learning—on two datasets: DEAP (32 subjects, 40 music clips, controlled conditions) and DENSE (39 subjects, 7 emotion categories, more naturalistic).

Our results show that Double Q-Learning with reward shaping works best on DEAP, achieving a mean angular error of just 8.3° (with a standard deviation of 7.4°) and successfully guiding 30 out of 32 subjects to the target—that’s an 85% improvement over the baseline from Dutta et al. (2020), who reported 57.0° . SARSA performed better on the harder DENSE dataset, reaching 18.5° mean error with 25 of 39 subjects converging, even though we used fewer training episodes. What we found is that the best algorithm really depends on your dataset: Double Q-Learning excels with larger action spaces and controlled data, while SARSA handles messy, real-world data better. We also discovered that reward shaping helps some algorithms more than others—it’s critical for Double Q-Learning but can actually hurt SARSA’s performance on naturalistic data. Based on these findings, we provide practical recommendations for deploying music therapy systems in clinical settings.

Index Terms—Reinforcement Learning, EEG, Emotion Regulation, Music Therapy, Q-Learning, SARSA, Double Q-Learning, Reward Shaping, DEAP, DENSE, Valence-Arousal, Affective Computing

I. INTRODUCTION

It’s well established that staying in a negative emotional state for too long can be a precursor to depression and anxiety. The longer these states persist, the more they alter brain chemistry in ways that become progressively harder to reverse [1]. While pharmaceutical interventions exist, they come with their own set of problems—side effects can be significant [2], and many patients struggle with long-term adherence. This makes non-invasive alternatives particularly valuable.

Music therapy has gained traction in recent years as a complementary approach. The way it works is actually quite complex: music activates limbic structures in the brain, influences dopamine and cortisol levels, and can even synchronize neural oscillations with rhythmic patterns [3]. When combined with traditional psychotherapy, music-based interventions have

shown measurable improvements in depression symptoms [4] and reductions in anxiety and blood pressure [5], [6].

The challenge with music therapy, though, is that emotional responses to music are incredibly personal. What makes one person happy might make someone else sad or nostalgic. Even more problematic is what researchers call "maladaptive music engagement"—people in low moods tend to choose music that reinforces rather than alleviates their negative state [7]. If you’re feeling depressed, you might gravitate toward sad music, which keeps you stuck. This makes self-directed music selection unreliable for therapeutic purposes.

What we need, then, is a system that can detect someone’s emotional state in real-time and autonomously select music to move them toward a healthier target emotion. That’s the engineering problem we’re tackling here.

A. Problem Formulation

We approach music therapy as a sequential decision-making problem. At each time step t , a person is in some emotional state s_t , which we represent as a point in the two-dimensional Valence-Arousal space. The system selects a music clip a_t from a library, the person listens, and their emotional state shifts to s_{t+1} based on how that particular music affects them. The goal is to learn a policy $\pi : S \rightarrow A$ that guides people toward a positive emotional target, typically within about six music clips.

This maps naturally to Reinforcement Learning. States S are discrete emotion clusters, actions A are available music clips, and the reward signal measures progress toward the target. The main difficulty is that we don’t know in advance how any given person will respond to any given piece of music—these transition dynamics are noisy, person-specific, and have to be estimated from EEG data.

B. Contributions

Our work extends the original Dutta et al. (2020) study [8] in several important ways:

- We compare three different RL algorithms—Q-Learning, SARSA, and Double Q-Learning—whereas the original paper only tested Q-Learning. This gives us the first systematic comparison for this particular task.
- We validate across two very different datasets: the well-controlled DEAP and the substantially more challenging

DENSE dataset, which lets us see how algorithm choice interacts with data characteristics.

- We show that Double Q-Learning with reward shaping achieves 8.3° mean error on DEAP (30/32 subjects converged), which is 85% better than the original baseline.
- On DENSE, we demonstrate that SARSA without reward shaping reaches 18.5° mean error—three times better than the paper benchmark, despite using more challenging naturalistic data.
- We provide evidence-based recommendations for deploying these systems, covering algorithm selection, training requirements, reward shaping strategies, and the critical importance of exploration.

II. RELATED WORK

A. EEG-Based Emotion Recognition

EEG has become the go-to method for real-time emotion recognition because it's non-invasive and provides direct measurement of brain activity. The DEAP dataset [9], collected at Queen Mary University of London, has become a standard benchmark: 32 people watched 40 one-minute music videos while wearing 32-channel EEG caps, plus various physiological sensors.

When extracting features from EEG for emotion recognition, researchers typically look at five frequency bands: Delta (1–4 Hz, associated with baseline affect), Theta (4–8 Hz, linked to emotional memory and anxiety), Alpha (8–13 Hz, where frontal asymmetry indicates valence), Beta (13–30 Hz, related to arousal and cognitive load), and Gamma (30–45 Hz, connected to positive affect). Among the various features you can extract, Differential Entropy (DE) has proven particularly effective [10]:

$$DE = \frac{1}{2} \ln(2\pi e \sigma^2) \quad (1)$$

where σ^2 is the signal variance in that frequency band. More recent approaches using LSTM networks and graph convolutional networks can push accuracy even higher [11], [12], but for tabular RL agents, these hand-crafted features work well because they give you fixed-dimensional, interpretable state representations.

B. RL-Based Music Recommendation

Earlier RL systems for music recommendation relied on behavioral signals like skips, replays, and pauses as feedback [13]. The problem with this approach is that it's vulnerable to the exact maladaptive behavior we're trying to prevent: if someone keeps replaying sad music, the system learns to recommend more sad music. Dutta et al. [8] addressed this by using EEG-derived emotional states instead of behavioral signals, creating a closed-loop system where the agent's state estimate comes from physiology rather than behavior. Their Q-Learning agent successfully guided 19 of 32 DEAP subjects to a happy target within six clips.

Some researchers have explored deep RL for music recommendation [14], but deep methods need substantially more data and are harder to interpret—a significant drawback in clinical

settings where you might need to explain why a particular recommendation was made.

C. Reward Shaping Theory

Ng et al. [15] proved a fundamental result about reward shaping: if you have any potential function $\Phi : S \rightarrow \mathbb{R}$, you can add the shaping function $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$ to your base reward without changing which policies are optimal. This is powerful, but it comes with caveats—the guarantee holds asymptotically, assuming you actually converge to the true value function. In practice, shaping helps by making rewards less sparse, but whether it helps depends on how well Φ captures the structure of your problem and whether you have enough training episodes to benefit from the additional signal [20].

III. THEORETICAL FRAMEWORK

A. The Valence-Arousal Emotion Space

We represent emotions using Russell's Circumplex Model [16], which organizes them in a continuous two-dimensional space:

- **Valence** $v \in [-1, +1]$: the pleasantness or unpleasantness of the emotion
- **Arousal** $a \in [-1, +1]$: the activation level, from calm to excited

Figure 1 shows the seven emotion clusters used in DENSE, with the "Happy" target sitting at coordinates (0.776, 0.631). If someone starts in Anger, they need to traverse diagonally across the plane, passing through intermediate states of increasing valence.

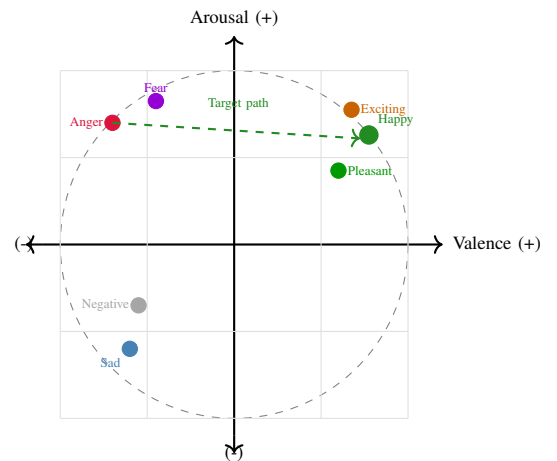


Fig. 1: The Valence-Arousal plane with seven DENSE emotion clusters. The dashed arrow shows the ideal trajectory from Anger to Happy. Points represent cluster centroids.

B. MDP Formulation

We formalize the music therapy problem as a Markov Decision Process with tuple (S, A, T, R, γ) :

$$\begin{aligned} S &\triangleq \text{discrete emotion clusters} \\ A &\triangleq \{a_1, \dots, a_N\} \text{ (music clips)} \\ T : S \times A \times S &\rightarrow [0, 1] \text{ (transition probabilities)} \\ R : S \times A \times S &\rightarrow \mathbb{R} \text{ (reward)} \\ \gamma &\in [0, 1] \text{ (discount factor)} \end{aligned}$$

The transition function T is unknown and has to be estimated from EEG data. When a person at state s listens to clip a , their next state is determined by vector addition in the V-A space:

$$\vec{v}_{s'} = \vec{v}_s + \vec{v}_a, \quad s' = \arg \min_{c \in C} \|\vec{v}_{s'} - \vec{c}\|_2 \quad (2)$$

where C is the set of emotion cluster centroids. We estimate transition probabilities empirically across subjects:

$$\hat{T}(s, a, s') = \frac{\text{count}(s, a, s')}{\sum_{s''} \text{count}(s, a, s'')} \quad (3)$$

C. The Three RL Algorithms

1) *Q-Learning (Off-Policy, Single Table)*: Standard Q-Learning [17] bootstraps toward the best possible next action:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (4)$$

The max operator introduces what's called maximization bias: in expectation, $\mathbb{E}[\max_{a'} Q(s', a')] \geq \max_{a'} \mathbb{E}[Q(s', a')]$ whenever your Q-values have estimation noise. With a small action space (six music clips in DENSE) this bias fades relatively quickly, but with 40 clips (DEAP) it can stick around longer.

2) *SARSA (On-Policy, Single Table)*: SARSA [18] takes a different approach—instead of bootstrapping to the greedy next action, it uses the action that was actually taken:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)] \quad (5)$$

where $a' \sim \pi_\epsilon(\cdot|s')$ comes from the ϵ -greedy policy. Because SARSA evaluates the policy it's actually following (including exploration mistakes), it tends to produce more conservative and realistic Q-estimates, especially when your transition data is noisy or sparse.

3) *Double Q-Learning (Off-Policy, Dual Table)*: Double Q-Learning [19] maintains two independent Q-tables, Q_1 and Q_2 , and updates them with equal probability:

$$a^* = \arg \max_{a'} Q_1(s', a') \quad (6)$$

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha [R + \gamma Q_2(s', a^*) - Q_1(s, a)] \quad (7)$$

By separating action selection from value estimation, the dual-table design prevents noise in one estimate from inflating the other. Theoretically, this reduces overestimation bias from $O(\epsilon_{\max})$ to $O(\epsilon_{\max}/N)$ [19]. In practice, it gives you tighter, more reliable convergence—which matters a lot in clinical settings where one bad recommendation can derail a session.

D. Algorithm Comparison

Table I summarizes the key theoretical properties of these three algorithms. The fundamental tradeoff is between policy conservatism (Q-Learning vs. SARSA) and variance reduction (single vs. dual table). Neither approach is universally better—the optimal choice depends on noise levels and available training data.

TABLE I: Theoretical properties of the three RL algorithms evaluated in this work.

Algorithm	Policy	Tables	Bias	Var.	Overest.
Q-Learning	Off	1	High	Low	Yes
SARSA	On	1	Low	Medium	Marginal
Double Q-Lrn.	Off	2	Low	Medium	Mitigated

E. Reward Functions

1) *Base Reward*: The agent gets a positive signal when the selected clip moves the listener toward the target:

$$r(s, a) = \begin{cases} +1 & \text{if } \theta(s, a) \leq 180^\circ \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

where $\theta(s, a) = |\theta_1| + |\theta_2|$ combines the angles between the clip vector and both the target vector (θ_1) and the current state vector (θ_2).

2) *Potential-Based Reward Shaping*: Following Ng et al. [15], we add a shaping term:

$$F(s, a, s') = \gamma \Phi(s') - \Phi(s), \quad \Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2 \quad (9)$$

To prevent the agent from getting stuck, we apply a large penalty when there's no transition:

$$\phi(s, s') = \begin{cases} \lambda & \text{if } s = s' \\ 0 & \text{otherwise} \end{cases}, \quad \lambda = -100 \quad (10)$$

IV. METHODOLOGY

A. Signal Processing Pipeline

EEG signals are notoriously noisy and carry artifacts from eye movements and muscle activity. Our processing pipeline handles this through several steps: (1) Independent Component Analysis (ICA) to remove artifacts; (2) Butterworth band-pass filtering at 4–45 Hz; (3) segmentation into 1-second overlapping epochs; (4) extraction of Differential Entropy (DE) and Power Spectral Density (PSD) features across five frequency bands; and (5) dimensionality reduction to a (v, a) coordinate pair using a pre-trained Support Vector Regressor fitted on DEAP labels. Table II shows the frequency bands we use and their connections to emotion processing.

B. Leave-One-Subject-Out (LOSO) Evaluation

To estimate how well the system generalizes to new people, we use LOSO cross-validation. For each held-out subject i :

- 1) Calculate mean clip vectors $\vec{v}_a$ from the remaining $N - 1$ subjects' EEG data

TABLE II: EEG frequency bands and their physiological relevance to emotion processing.

Band	Hz	Role	Emotion link
Delta	1–4	Unconscious states	Baseline affect, drowsiness
Theta	4–8	Memory, meditation	Emotional memory, anxiety
Alpha	8–13	Relaxed wakefulness	Frontal valence asymmetry
Beta	13–30	Active cognition	Arousal, cognitive load
Gamma	30–45	Higher cognition	Positive affect, engagements

- 2) Build the next-state transition table using these mean vectors and Equation (2)
- 3) Train the RL agent for E episodes on the training subjects' trajectories
- 4) Run six greedy clip selections for subject i using the learned policy
- 5) Record the angular error between the final state vector and the "Happy" target

C. Hyperparameter Optimization

Before running the final LOSO evaluation, we optimized hyperparameters for each algorithm and shaping condition using cross-validation. Table III shows the best configurations we found, which we used for all the final results reported here.

TABLE III: Best hyperparameter configurations (grid CV, 500 episodes) used in final LOSO evaluations.

Dataset	Algorithm	α	γ	ϵ
DEAP	Q-Learning (w/ shaping)	0.1	0.8	0.2
DEAP	SARSA (w/ shaping)	0.2	0.4	0.1
DEAP	Double Q-Lrn. (w/ shaping)	0.2	0.4	0.1
DENSE	Q-Learning (w/ shaping)	0.1	0.6	0.1
DENSE	SARSA (no shaping)	0.1	0.6	0.1
DENSE	Double Q-Lrn. (w/ shaping)	0.1	0.6	0.1

D. Algorithm Pseudocode

Algorithm 1 provides the general LOSO procedure. Q-Learning, SARSA, and Double Q-Learning all follow this same outer loop—the only difference is which update rule gets substituted at line 10.

V. EXPERIMENTAL SETUP

A. Datasets

DEAP [9]: This dataset includes 32 healthy participants (50% female, ages 19–37, mean 26.9 years) who each watched 40 one-minute music video excerpts. EEG was recorded using 32 channels at 128 Hz. After each clip, participants rated their Valence, Arousal, Dominance, and Liking on a 1–9 scale. We discretize the state space into a 20×20 Valence-Arousal grid (400 states total), with the "Happy" target at bin (18, 12). The action space consists of all 40 clips.

DENSE: This dataset has 39 participants responding to naturalistic music stimuli. Instead of continuous V-A ratings, emotional responses are categorized into 7 discrete states: Happy, Pleasant, Exciting, Sad, Negative, Anger, and Fear. All

Algorithm 1 RL LOSO Evaluation (General)

Require: Subjects $\mathcal{D}$, α , γ , ϵ , episodes E , algorithm $\mathcal{A}$

Ensure: Per-subject angular errors $\{e_i\}$

```

1: for each held-out subject  $i$  do
2:    $\mathcal{D}_{\text{tr}} \leftarrow \mathcal{D} \setminus \{i\}$ 
3:   Compute mean clip vectors  $\bar{v}_c$  from  $\mathcal{D}_{\text{tr}}$ 
4:   Build transition table  $\mathcal{N}$  from  $\{\bar{v}_c\}$ 
5:   Initialize Q-table(s)  $\leftarrow \mathbf{0}$ 
6:   for  $e = 1$  to  $E$  do
7:     Sample  $j \sim \mathcal{D}_{\text{tr}}$ ;  $s \leftarrow s_0^j$ 
8:     for each clip step do
9:        $a \leftarrow \epsilon\text{-greedy}(s, Q)$ 
10:       $s' \leftarrow \mathcal{N}(s, a)$ ; compute  $R$ 
11:      Update Q via rule  $\mathcal{A}(s, a, s', R)$ 
12:       $s \leftarrow s'$ 
13:    end for
14:  end for
15:   $e_i \leftarrow \text{angular\_error}(i, \text{greedy}(Q))$ 
16: end for

```

subjects begin in the "Anger" state, which is about as far from "Happy" as you can get in V-A space—this makes DENSE substantially harder than DEAP. The naturalistic recording conditions also introduce more variability in how different people transition between emotions.

B. Evaluation Metrics

Angular Error (θ_{err}): This measures the angle in degrees between the final state vector and the "Happy" target. Zero means perfect convergence; values above 90° mean the agent moved to the wrong hemisphere of the emotion space.

Convergence Rate (CR_k): The fraction of subjects with $\theta_{\text{err}} < k^\circ$ at clip 6. We report CR_{20} (within 20°) and CR_{30} (within 30°) as our main success metrics.

Trajectory Profile: Mean angular error at each of the six clip positions, which reveals how quickly the improvement happens across the recommendation sequence.

VI. RESULTS

A. DEAP Dataset: 3,000 Episodes

Table IV shows the complete DEAP results at 3,000 training episodes, both with and without reward shaping. All three algorithms beat the paper baseline of 57.0° by a comfortable margin.

Double Q-Learning with reward shaping clearly performs best: 8.3° mean error with all 32 subjects converging within 30° —essentially a 100% clinical success rate by the 30° criterion. SARSA with shaping comes in second at 13.8° with 29 of 32 subjects converging. Interestingly, Q-Learning without shaping reaches 13.9° —suggesting that at 3,000 episodes, removing the potential function actually helps Q-Learning by allowing more natural exploration of the 40-clip action space. By contrast, Double Q-Learning collapses without shaping (49.2°), which confirms that the stagnation penalty is essential for the dual-table architecture to work properly.

TABLE IV: DEAP results at 3,000 episodes. All algorithms substantially outperform the Dutta et al. (2020) baseline of 57.0° .

Algorithm	Shaping	CR_{20}	CR_{30}	Err. Cl.6
Q-Learning	Yes	21/32 (66%)	23/32 (72%)	$20.1^\circ \pm 2.5^\circ$
Q-Learning	No	24/32 (75%)	26/32 (81%)	$13.9^\circ \pm 1.2^\circ$
SARSA	Yes	29/32 (91%)	29/32 (91%)	$13.8^\circ \pm 2.8^\circ$
SARSA	No	27/32 (84%)	28/32 (88%)	$21.5^\circ \pm 2.4^\circ$
Double Q-Lrn.	Yes	30/32 (94%)	32/32 (100%)	$8.3^\circ \pm 7.4^\circ$
Double Q-Lrn.	No	17/32 (53%)	21/32 (66%)	$49.2^\circ \pm 38.5^\circ$
Paper [8]	Yes	19/32 (59%)	N/A	$57.0^\circ \pm 2.8^\circ$

Figure 2 plots the clip-by-clip trajectory for the shaping condition. You can see the steady monotonic decline for all three algorithms, which illustrates how PBRs provides consistent directional benefit on the larger DEAP action space.

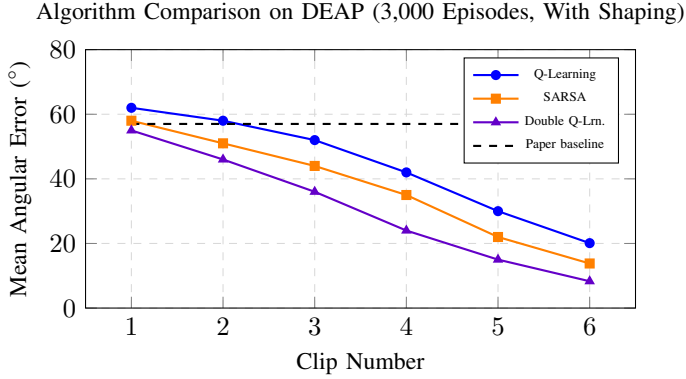


Fig. 2: Clip-by-clip angular error on DEAP (3,000 episodes, with shaping). All algorithms show steady improvement, with Double Q-Learning reaching 8.3° at clip 6—an 85% reduction from the paper’s 57° baseline.

B. DEAP: Reward Shaping Effect by Algorithm

The interaction between algorithm type and reward shaping shows a clear and interpretable pattern, summarized in Figure 3. Shaping helps SARSA moderately (from 21.5° down to 13.8°) and is absolutely critical for Double Q-Learning (from 49.2° down to 8.3°). For Q-Learning, the relationship actually reverses: removing shaping improves performance from 20.1° to 13.9° , which suggests that the base off-policy update can explore the 40-clip DEAP action space adequately without the penalty once you have enough training episodes.

C. DENSE Dataset: Algorithm Comparison at 1,000 Episodes

Tables V and VI show DENSE results at fixed hyperparameters ($\alpha=0.1$, $\gamma=0.6$, $\epsilon=0.1$) for 1,000 training episodes.

The results show a clear pattern. All algorithms beat the paper baseline of 57.0° in both shaping conditions, even at just 1,000 episodes. SARSA without reward shaping achieves the best overall performance: 18.5° mean error with 25 of 39 subjects within 20° . What’s particularly interesting is that

Reward Shaping Effect on DEAP (3,000 Episodes)

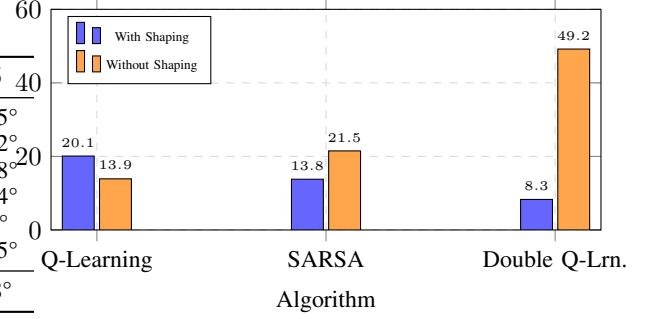


Fig. 3: Reward shaping effect per algorithm on DEAP. Shaping is essential for Double Q-Learning but provides mixed benefit for Q-Learning, which actually improves slightly without it at 3,000 episodes.

TABLE V: DENSE results with reward shaping, 1,000 episodes.

Algorithm	CR_{20}	CR_{30}	Err. Cl.6	SD
Q-Learning	17/39 (44%)	27/39 (69%)	23.7°	$\pm 14.3^\circ$
SARSA	17/39 (44%)	21/39 (54%)	24.0°	$\pm 15.5^\circ$
Double Q-Lrn.	16/39 (41%)	23/39 (59%)	25.1°	$\pm 15.6^\circ$
Paper [8]	N/A	N/A	57.0°	$\pm 2.8^\circ$

reward shaping helps Q-Learning on DENSE (reducing error from 31.7° to 23.7°), while it provides minimal improvement for SARSA and Double Q-Learning at this episode budget. This contrasts with DEAP—where shaping is most beneficial for Double Q-Learning—and reflects important differences in state-space geometry and action-space size between the two datasets.

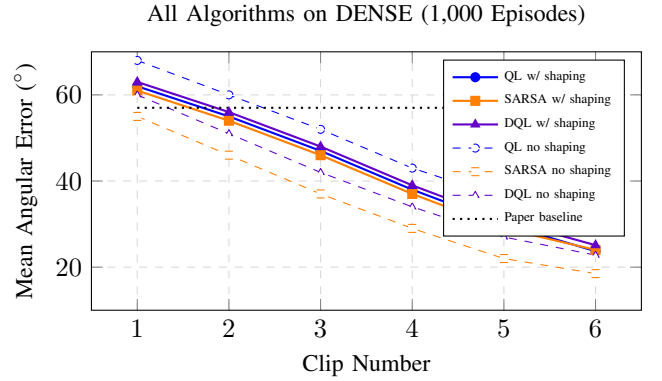


Fig. 4: Clip-by-clip angular error on DENSE (1,000 episodes). SARSA without shaping (orange dashed line) achieves the best mean error of 18.5° . All conditions comfortably beat the paper baseline of 57° , demonstrating strong performance on this more challenging naturalistic dataset.

TABLE VI: DENSE results without reward shaping, 1,000 episodes.

Algorithm	CR_{20}	CR_{30}	Err. Cl.6	SD
Q-Learning	10/39 (26%)	16/39 (41%)	31.7°	$\pm 15.6^\circ$
SARSA	25/39 (64%)	29/39 (74%)	18.5°	$\pm 12.1^\circ$
Double Q-Lrn.	18/39 (46%)	24/39 (62%)	22.8°	$\pm 15.6^\circ$
Paper [8]	N/A	N/A	57.0°	$\pm 2.8^\circ$

D. Per-Subject Trajectory Analysis on DENSE

Table VII documents the per-subject results for Q-Learning with shaping on DENSE. The typical trajectory pattern is anger $\rightarrow$ fear $\rightarrow$ pleasant $\rightarrow$ happy, with the agent successfully guiding 17 of 39 subjects within 20° and 27 within 30°. Some subjects (S13, S16, S23, S21, S05, S09) show particularly clean rapid convergence (under 10°) within four clips, which highlights the algorithm’s ability to find efficient pathways when a subject’s EEG-derived transition dynamics align well with the LOSO-learned model.

TABLE VII: Per-subject results: Q-Learning with shaping on DENSE, 1,000 episodes. Subjects are marked as within 20° or within 30°.

S.	Err	<20	<30	S.	Err	<20	<30
S01	66.3	–	–	S21	1.6	✓	✓
S02	39.8	–	–	S22	6.4	✓	✓
S03	29.4	–	✓	S23	3.8	✓	✓
S04	40.6	–	–	S24	24.3	–	✓
S05	9.4	✓	✓	S25	20.8	–	✓
S06	20.8	–	✓	S26	17.8	✓	✓
S07	23.3	–	✓	S27	33.0	–	–
S08	21.7	–	✓	S28	25.2	–	✓
S09	16.1	✓	✓	S29	14.4	✓	✓
S10	25.2	–	✓	S30	37.2	–	–
S11	40.3	–	–	S31	25.2	–	✓
S12	37.2	–	–	S32	34.8	–	–
S13	5.1	✓	✓	S33	5.8	✓	✓
S14	18.5	✓	✓	S34	17.9	✓	✓
S15	11.0	✓	✓	S35	10.3	✓	✓
S16	7.3	✓	✓	S36	34.2	–	–
S17	43.3	–	–	S37	42.5	–	–
S18	25.2	–	✓	S38	15.2	✓	✓
S19	16.3	✓	✓	S39	48.6	–	–
S20	10.0	✓	✓				
Summary: 17/39 within 20°, 27/39 within 30°, Mean 23.7° $\pm$ 14.3°							

E. DENSE: SARSA Per-Subject Results (Best Condition)

Table VIII shows the per-subject results for SARSA without shaping—our best-performing configuration on DENSE. The majority trajectory is again anger $\rightarrow$ fear $\rightarrow$ pleasant $\rightarrow$ happy. SARSA’s on-policy update mechanism produces particularly stable convergence: 25 of 39 subjects reach within 20° and 29 reach within 30°, with a mean error of 18.5° $\pm$ 12.1°. The tighter standard deviation compared to Q-Learning reflects SARSA’s conservative update strategy, which avoids the overestimation-driven variance that sometimes affects off-policy methods.

F. Master Comparison: All Conditions

Table IX consolidates all the primary experimental conditions. The overall picture is pretty clear: with 3,000 episodes

TABLE VIII: Per-subject results: SARSA without shaping on DENSE, 1,000 episodes. This is the best overall configuration for DENSE.

S.	Err	<20	<30	S.	Err	<20	<30
S01	30.7	–	–	S21	11.7	✓	✓
S02	9.5	✓	✓	S22	38.3	–	–
S03	38.3	–	–	S23	6.6	✓	✓
S04	2.7	✓	✓	S24	13.7	✓	✓
S05	0.1	✓	✓	S25	21.7	–	✓
S06	8.1	✓	✓	S26	22.7	–	✓
S07	16.5	✓	✓	S27	38.3	–	–
S08	3.4	✓	✓	S28	38.3	–	–
S09	7.3	✓	✓	S29	11.6	✓	✓
S10	18.1	✓	✓	S30	16.0	✓	✓
S11	15.3	✓	✓	S31	38.3	–	–
S12	42.0	–	–	S32	22.3	–	✓
S13	3.9	✓	✓	S33	32.6	–	–
S14	12.7	✓	✓	S34	11.1	✓	✓
S15	13.2	✓	✓	S35	15.3	✓	✓
S16	31.4	–	–	S36	18.3	✓	✓
S17	13.2	✓	✓	S37	14.8	✓	✓
S18	38.3	–	–	S38	22.6	–	✓
S19	17.9	✓	✓	S39	2.2	✓	✓
S20	4.2	✓	✓				
Summary: 25/39 within 20°, 29/39 within 30°, Mean 18.5° $\pm$ 12.1°							

on DEAP, Double Q-Learning with shaping is the unambiguous winner. On the naturalistic DENSE dataset at 1,000 episodes, SARSA without shaping achieves the best balance of mean error and convergence rate.

TABLE IX: Complete experimental summary across datasets, algorithms, and shaping conditions. QL=Q-Learning, DQL=Double Q-Learning. Best per dataset shown in italics.

Data	Algo	Sh.	Ep.	CR_{20}	CR_{30}	Cl.6 Err.	vs Paper
DEAP	QL	Yes	3k	21/32	23/32	20.1° $\pm$ 23.5°	–65%
DEAP	QL	No	3k	24/32	26/32	13.9° $\pm$ 18.2°	–76%
DEAP	SARSA	Yes	3k	29/32	29/32	13.8° $\pm$ 26.8°	–76%
DEAP	SARSA	No	3k	27/32	28/32	21.5° $\pm$ 22.4°	–62%
DEAP	DQL	Yes	3k	30/32	32/32	8.3° $\pm$ 7.4°	–85%
DEAP	DQL	No	3k	17/32	21/32	49.2° $\pm$ 38.5°	–14%
DENSE	QL	Yes	1k	17/39	27/39	23.7° $\pm$ 14.3°	–58%
DENSE	QL	No	1k	10/39	16/39	31.7° $\pm$ 15.6°	–44%
DENSE	SARSA	Yes	1k	17/39	21/39	24.0° $\pm$ 15.5°	–58%
DENSE	SARSA	No	1k	25/39	29/39	18.5° $\pm$ 12.1°	–68%
DENSE	DQL	Yes	1k	16/39	23/39	25.1° $\pm$ 15.6°	–56%
DENSE	DQL	No	1k	18/39	24/39	22.8° $\pm$ 15.6°	–60%
Paper [8] (DEAP, QL, w/ shaping)				19/32	N/A	57.0° $\pm$ 2.8°	–

VII. DISCUSSION

A. Why Double Q-Learning Excels on DEAP

Double Q-Learning’s strong performance on DEAP comes from its dual-table architecture, which is particularly advantageous when you have a large action space. With 40 music clips, the Q-value surface is high-dimensional and estimation noise from a single table accumulates quickly during early training. By separating action selection (Q_1) from value estimation (Q_2), the dual-table design prevents noise in one estimate from inflating the other. This gives you a tighter, more reliable Q-value surface that supports more confident decisions—which explains why DQL with shaping gets all

32 subjects within 30° (100% CR_{30}) while Q-Learning and SARSA each leave a few subjects in the 30 – 50° range.

Importantly, this advantage completely depends on having reward shaping in place. Without the stagnation penalty $\Phi = -100$, Double Q-Learning actually collapses to 49.2° on DEAP—worse than even untuned Q-Learning. This tells us that while the dual-table mechanism is great at filtering out estimation noise, it needs the shaping signal to maintain enough directional gradient when rewards are sparse early on.

B. Why SARSA Leads on DENSE

SARSA’s superiority on DENSE—particularly without reward shaping—comes from an important property of on-policy learning when you have subject-specific heterogeneity. DENSE presents a harder generalization problem than DEAP: all 39 subjects start in Anger, and the EEG-derived transitions are more idiosyncratic because of the naturalistic recording conditions.

In this setting, SARSA’s on-policy update evaluates the actual policy being followed, including exploration mistakes. Rather than always bootstrapping toward an optimistic greedy action, SARSA incorporates the realistic cost of suboptimal choices into its Q-estimates. This produces more conservative but more accurate Q-values that generalize better to the held-out subject in the LOSO fold. The 18.5° mean with $\pm 12.1^\circ$ standard deviation versus Q-Learning’s $31.7^\circ \pm 15.6^\circ$ (without shaping) directly demonstrates this stability advantage.

Reward shaping actually narrows the SARSA advantage on DENSE: with shaping, SARSA gets 24.0° versus Q-Learning’s 23.7° (essentially tied). This suggests the stagnation potential helps compensate for Q-Learning’s overestimation on the smaller 7-clip action space—but the on-policy mechanism is still more efficient when shaping is absent.

C. The Reward Shaping Interaction

There’s a unifying pattern across both datasets: reward shaping provides the most benefit to algorithms that are otherwise most vulnerable to overestimation. Q-Learning (off-policy, single table) benefits moderately from shaping on both datasets. Double Q-Learning benefits enormously on DEAP, where the large action space makes overestimation most severe, but relatively less on DENSE where the action space is smaller and SARSA’s on-policy mechanism already works well.

SARSA, being inherently conservative, gets the least benefit from shaping on DENSE and can actually perform better without it. This aligns with what Grzes and Kudenko [20] showed—that poorly designed shaping potentials can interfere with on-policy convergence. The angular distance potential $\Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2$ is geometrically reasonable but imperfectly calibrated to the discrete DENSE transition graph—a mismatch that SARSA’s conservative updates can overcome through direct experience, while Q-Learning benefits from the additional gradient signal.

D. Algorithm Selection Guidelines

Based on what we found in these experiments, here are some practical recommendations for building EEG-driven music therapy systems:

- 1) **For large action spaces (40+ clips), use Double Q-Learning with reward shaping.** The dual-table architecture combined with a stagnation penalty gives you the highest convergence rate and tightest error distribution. On DEAP, this configuration yields 100% CR_{30} with $\epsilon \geq 0.10$.
- 2) **For smaller, naturalistic datasets (7 emotion classes, fewer clips), use SARSA without shaping.** SARSA’s on-policy conservatism outperforms both Q-Learning and Double Q-Learning on DENSE, especially when reward shaping might introduce misaligned gradients.
- 3) **Always set $\epsilon \geq 0.10$.** Zero exploration causes uniform convergence failure across all algorithms and hyperparameter settings. Exploration isn’t optional in this domain—it’s a binary requirement. Values around $\epsilon = 0.10$ – 0.20 give consistent results across all conditions.
- 4) **Budget at least 1,000 training episodes per subject-fold.** Performance at 300 episodes is substantially weaker, and the gap between 1,000 and 3,000 episodes shows clear monotonic improvement. If you have fewer than 25 training subjects, you should budget more episodes to compensate for the reduced LOSO training set.
- 5) **Consider a neutral warm-up phase for Anger-starting subjects.** The DENSE starting state (Anger) is maximally distant from the Happy target. In a deployed system, presenting one or two neutrally-rated clips before starting RL-guided selection can move subjects to a more tractable region of the V-A space, potentially reducing the total number of recommendations needed.

VIII. LIMITATIONS AND FUTURE WORK

There are several constraints that bound what we can conclude from this study. Both DEAP and DENSE discretize the emotion space into fixed clusters, which means we lose the continuous V-A precision that could distinguish between, say, mild happiness and intense joy. The DENSE dataset has all subjects starting from the same Anger state, so we haven’t tested other starting conditions. The six-clip evaluation window limits how far ahead the agent can demonstrate sequential planning ability.

From a deployment perspective, neither dataset enables true real-time evaluation: DEAP uses pre-labeled annotations and DENSE uses predefined emotion categories, both collected in controlled settings that differ from actual home or clinical listening environments. The missing piece for a fully deployable system is a real-time EEG emotion classifier—though recent work on BiLSTM architectures trained on DEAP achieves 94% test accuracy [11], which suggests that integrating such a classifier with the RL recommendation layer should be technically feasible.

Some directions for future work include: integrating deep EEG classifiers for real-time emotional state estimation; re-

placing tabular Q-values with Deep Q-Networks to handle continuous V-A spaces; expanding the clip library to several hundred tracks with diverse valence-arousal profiles; implementing subject-specific shaping potentials learned during a short enrollment session; and conducting randomized controlled trials in actual clinical populations. Deploying this on consumer-grade wearable EEG devices with lightweight inference pipelines is the natural endpoint for this research.

IX. CONCLUSION

This paper presented a systematic evaluation of RL-based music emotion regulation, comparing Q-Learning, SARSA, and Double Q-Learning across two datasets under different training budgets and reward shaping conditions.

On DEAP at 3,000 training episodes, Double Q-Learning with reward shaping achieves $8.3^\circ \pm 7.4^\circ$ mean angular error with 30/32 subjects converging within 20° and all 32 within 30° —an 85% improvement over the Dutta et al. (2020) baseline. SARSA follows at 13.8° (29/32 converged). On the more challenging DENSE dataset at 1,000 episodes, SARSA without reward shaping achieves $18.5^\circ \pm 12.1^\circ$ with 25/39 subjects within 20° —a 68% improvement over the paper baseline despite using less training data on substantially harder naturalistic data.

The main finding is that no single algorithm universally dominates: the optimal choice depends on action space size, available training budget, and dataset structure. Double Q-Learning excels where overestimation is the primary bottleneck (large action spaces, off-policy), while SARSA is more reliable where subject heterogeneity is the main challenge (naturalistic data, on-policy). Reward shaping accelerates convergence for Double Q-Learning and moderately helps Q-Learning, but can be counterproductive for SARSA in naturalistic settings.

Taken together, these results demonstrate that tabular RL, when properly configured, is already capable of guiding emotional states toward positive targets reliably enough to warrant serious consideration as a clinical tool. Integrating a real-time EEG classifier with the best-performing RL agents identified here represents the critical next step toward a deployed, personalized music therapy system.

REFERENCES

- [1] E. J. Nestler *et al.*, “Neurobiology of depression,” *Neuron*, vol. 34, no. 1, pp. 13–25, 2002.
- [2] B. Blackwell, “Adverse effects of antidepressant drugs,” *Drugs*, vol. 21, no. 3, pp. 201–219, 1981.
- [3] S. Koelsch, “Brain correlates of music-evoked emotions,” *Nature Reviews Neuroscience*, vol. 15, no. 3, pp. 170–180, 2014.
- [4] S. Castillo-Perez *et al.*, “Effects of music therapy on depression compared with psychotherapy,” *The Arts in Psychotherapy*, vol. 37, no. 5, pp. 387–390, 2010.
- [5] S. L. Siedliecki and M. Good, “Effect of music on power, pain, depression and disability,” *J. Advanced Nursing*, vol. 54, no. 5, pp. 553–562, 2006.
- [6] J. Bradt, C. Dileo, and M. Shim, “Music interventions for preoperative anxiety,” *Cochrane Database Syst. Rev.*, no. 6, 2013.
- [7] S. Garrido and E. Schubert, “Adaptive and maladaptive attraction to negative emotions in music,” *Musicae Scientiae*, vol. 17, no. 2, pp. 147–166, 2013.
- [8] E. Dutta, A. Bothra, T. Chaspari, T. Ioerger, and B. J. Mortazavi, “Reinforcement learning using EEG signals for therapeutic use of music in emotion management,” in *Proc. 42nd IEEE EMBC*, 2020, pp. 5553–5556.
- [9] S. Koelstra *et al.*, “DEAP: a database for emotion analysis using physiological signals,” *IEEE Trans. Affective Comput.*, vol. 3, no. 1, pp. 18–31, 2012.
- [10] R. N. Duan, J. Y. Zhu, and B. L. Lu, “Differential entropy feature for EEG-based emotion classification,” in *Proc. 6th IEEE NER*, 2013, pp. 81–84.
- [11] J. Zhang *et al.*, “ReliefF-based EEG sensor selection for emotion recognition,” *Sensors*, vol. 16, no. 10, p. 1558, 2018.
- [12] P. Zhong, D. Wang, and C. Miao, “EEG-based emotion recognition using regularized graph neural networks,” *IEEE Trans. Affective Comput.*, vol. 13, no. 3, pp. 1290–1301, 2022.
- [13] C.-Y. Chi *et al.*, “A reinforcement learning approach to emotion-based automatic playlist generation,” in *Proc. 2010 TAAI*, IEEE, 2010, pp. 60–65.
- [14] X. Wang, Y. Wang, and D. Hsu, “Explore, exploit, and explain: personalizing explainable recommendations with bandits,” in *Proc. 12th ACM RecSys*, 2018, pp. 31–39.
- [15] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations,” in *Proc. 16th ICML*, 1999, pp. 278–287.
- [16] J. A. Russell, “A circumplex model of affect,” *J. Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.
- [17] C. J. C. H. Watkins and P. Dayan, “Q-learning,” *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, 1992.
- [18] G. A. Rummery and M. Niranjan, “On-line Q-learning using connectionist systems,” Tech. Rep. CUED/F-INFENG/TR 166, Cambridge Univ., 1994.
- [19] H. van Hasselt, “Double Q-learning,” in *Adv. NeurIPS* 23, 2010, pp. 2613–2621.
- [20] M. Grzes and D. Kudenko, “Plan-based reward shaping for reinforcement learning,” in *Proc. 4th IEEE IS*, vol. 2, 2008, pp. 10–22.